

Question 6: Investigative Task**4 points****General Scoring Notes**

- Each part of the question (indicated by a letter) is initially scored by determining if it meets the criteria for essentially correct (E), partially correct (P), or incorrect (I). The response is then categorized based on the scores assigned to each letter part and awarded an integer score between 0 and 4 (see the table at the end of the question).
- The model solution represents an ideal response to each part of the question, and the scoring criteria identify the specific components of the model solution that are used to determine the score.

Model Solution

- (a) (i) The relative frequencies of successful and unsuccessful treatments in each clinic are presented in the following table:

	Clinic A	Clinic B
Unsuccessful Treatment	$\frac{51}{139} \approx 0.3669$	$\frac{33}{68} \approx 0.4853$
Successful Treatment	$\frac{88}{139} \approx 0.6331$	$\frac{35}{68} \approx 0.5147$

- (ii) Clinic A appears to be more successful in treating allergy sufferers than Clinic B. Clinic A was successful for 63.3% of the allergy sufferers it treated, while Clinic B was successful for only 51.5% of the allergy sufferers it treated.

Scoring

Essentially correct (E) if the response satisfies the following two components:

- The response to part (a-i) provides correct numerical values for the relative frequencies in all four cells of the table
- The response to part (a-ii) identifies the clinic that is more successful in treating allergy sufferers and includes a justification based on the relative frequencies reported in part (a-i)

Partially correct (P) if the response satisfies only one of the two components
OR

if the response provides correct numerical values for the relative frequencies in only two or three of the cells of the table, and correctly identifies the clinic that has greater success, with or without providing adequate justification based on the values of the reported relative frequencies.

Incorrect (I) if the response does not meet the criteria for E or P.

Additional Notes:

- Responses may be given as proportions or percentages.
- To satisfy component 1, numerical values for relative frequencies must be accurately reported to at least two decimal places, e.g., 0.36 or 0.37, 0.48 or 0.49, 0.63 or 0.64, 0.51 or 0.52.
- If a response in part (a-i) reports the relative proportions of the total (dividing the count in each of the four cells by 207) instead of the conditional relative frequencies, then component 2 may be satisfied in either of the following ways:
 - Concluding that Clinic A is more successful in treating allergies based on the larger proportion of successful cases (0.43 vs. 0.17)
 - Concluding that Clinic B is more successful in treating allergies based on the smaller proportion of unsuccessful cases (0.16 vs. 0.25).

- If a response reports incorrect values in part (a-i), then the justification in component 2 may use either the relative frequencies reported in the table in part (a-i) OR the correct conditional relative frequencies given in the model solution.
 - To satisfy component 2, the response must make it clear that Clinic A is being compared to Clinic B, either by explicitly mentioning both clinics OR by reporting both relative frequencies that are being compared.
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Model Solution	Scoring
(b) No. This is an observational study; it is not a randomized experiment. Cause and effect can only be established with a well-designed, randomized experiment. There may be other variables, besides where a patient was treated, that affect the success rates for treating allergies. For example, Clinic A may mostly treat mild allergy cases that are easy to treat successfully, while Clinic B may mostly treat severe allergy cases that are more difficult to treat successfully.	Essentially correct (E) if the response satisfies the following two components: 1. Indicates “no,” a causal inference is not justified 2. Provides at least one of the following explanations: • Notes that this is an observational study • Indicates that this is not a randomized experiment • Identifies a plausible confounding variable that could affect the success rates for treating allergies at the two clinics Partially correct (P) if the response satisfies component 2 but fails to indicate that researchers are not justified in making a causal inference OR if the response satisfies component 1 AND includes one of the following as justification: • A statement that association does not imply causation • A statement that there may be a confounding variable without specifying a reasonable confounding variable. Incorrect (I) if the response does not satisfy the criteria for E or P.

Additional Notes:

- A response that states that treatments (Clinic A, Clinic B) are not assigned (randomly assigned, imposed) to the patients satisfies component 2.
- A response that identifies a reasonable confounding variable does not need to explain why it is a confounding variable in order to satisfy component 2.
- The strength of the response in part (b) should be considered if holistic scoring is necessary.

Model Solution	Scoring
(c) (i) Clinic A: More successful in treating mild allergies. Clinic A successfully treated 75.0% of mild allergy sufferers, while Clinic A successfully treated only 28.6% of severe allergy sufferers. Clinic B: More successful in treating mild allergies. Clinic B successfully treated 91.7% of mild allergy sufferers, while Clinic B successfully treated only 42.9% of severe allergy sufferers. (ii) Clinic A: More likely to treat mild allergy sufferers than severe allergy sufferers. Of the 139 allergy sufferers treated at Clinic A, 104 (74.8%) suffered from mild allergies, while only 35 (25.2%) suffered from severe allergies. Clinic B: More likely to treat severe allergy sufferers than mild allergy sufferers. Of the 68 allergy sufferers treated at Clinic B, 56 (82.4%) suffered from severe allergies, while only 12 (17.6%) suffered from mild allergies.	Essentially correct (E) if the response satisfies the following four components: 1. The response to part (c-i) indicates that both Clinic A and Clinic B are more successful in treating mild allergies than treating severe allergies 2. The response to part (c-i) provides a justification based on correctly reported proportions (or percentages) of successfully treated allergy sufferers of each severity for each clinic 3. The response to part (c-ii) indicates that Clinic A is more likely to treat mild allergy sufferers, while Clinic B is more likely to treat severe allergy sufferers 4. The response to part (c-ii) provides a justification based on correctly reported frequencies or proportions (or percentages) of severe or mild allergy sufferers treated by each clinic Note: See the Q6 Correct Relative Frequencies tables at the end of this scoring guideline for the relative frequencies that should be used in each designated part. Partially correct (P) if the response satisfies two or three of the four components. Incorrect (I) if the response does not meet the criteria for E or P.

Additional Notes:

- A response to part (c-i) may satisfy component 2 in either of the following ways:
 - For each clinic, reports the correct proportion (or percentage) of severe allergy sufferers successfully treated AND the correct proportion (or percentage) of mild allergy sufferers successfully treated.
 - Makes a correct reference to the mosaic plot, comparing the ratio of shaded area to open area within the “Mild” rectangle to the ratio of shaded area to open area within the “Severe” rectangle for each clinic. Numerical values are not required.
- A response to part (c-ii) may satisfy component 4 in any of the following ways:
 - For each clinic, reports the correct proportion (or percentage) of treated allergy sufferers with severe allergies. (Clinic A: 0.252, Clinic B: 0.824)
 - For each clinic, reports the correct proportion (or percentage) of treated allergy sufferers with mild allergies. (Clinic A: 0.748, Clinic B: 0.176)

- Reports the correct proportion (or percentage) of treated allergy sufferers with severe allergies for one clinic AND reports the correct proportion (or percentage) of treated allergy sufferers with mild allergies for the other clinic.
 - For each clinic, reports the correct frequency of treated allergy sufferers with mild allergies and the correct frequency of treated allergy sufferers with severe allergies. (Clinic A: 104 mild and 35 severe; Clinic B: 12 mild and 56 severe)
 - Makes a correct reference to the mosaic plot, comparing the area or width of the “Mild” rectangle to the area or width of the “Severe” rectangle within each clinic. Numerical values are not required.
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Model Solution	Scoring
(d) The more successful clinic identified in part (a-ii) is Clinic A, which is different from the physician's conclusion that Clinic B is better when taking allergy severity into account. This happens because for both clinics the success rate is much higher for mild allergy sufferers (75.0 percent versus 28.6 percent for Clinic A, and 91.7 percent versus 42.9 percent for Clinic B). Clinic A treats mostly mild allergy sufferers (74.8 percent of its patients) while Clinic B treats mostly severe allergy sufferers (82.4 percent of its patients). Therefore, when combining the results across allergy severity categories to obtain the table from part (a-i), the facts that Clinic A treats a larger proportion of mild allergy sufferers and mild allergy sufferers have a higher success rate make it appear as if Clinic A is better overall.	Essentially correct (E) if the response satisfies the following two components: 1. The response indicates that within each clinic, the proportion of successfully treated mild allergy sufferers is greater than the proportion of successfully treated severe allergy sufferers (0.75 vs. 0.286 for Clinic A and 0.917 vs. 0.429 for Clinic B) <i>OR</i> the response indicates that the overall proportion of successfully treated mild allergy sufferers (0.767) is much larger than the overall proportion of successfully treated severe allergy sufferers (0.374) 2. The response indicates that the allergy sufferers treated by Clinic A include a much greater proportion of mild cases than the allergy sufferers treated by Clinic B (sample proportions: 0.748 vs. 0.176) <i>OR</i> the response indicates that the allergy sufferers treated by Clinic B includes a much greater proportion of severe cases than the allergy patients treated by Clinic A (sample proportions: 0.824 vs 0.252) <i>OR</i> the response indicates that of the allergy sufferers treated at Clinic A, a higher proportion are mild (0.748) than severe (0.252), and of the allergy sufferers treated by Clinic B, a higher proportion are severe (0.824) than mild (0.176) Partially correct (P) if the response satisfies only one of the two components. Incorrect (I) if the response does not meet the criteria for E or P.

Additional Notes:

- A response that does not use the answer from (c-i) cannot satisfy component 1.
- A response that does not use the answer from (c-ii) cannot satisfy component 2.
- A response that compares successful treatments between clinics (0.75 vs. 0.917 or 0.286 vs. 0.429), does not satisfy component 1 or component 2.

- A response does not need to repeat the calculated values from part (c), but if it does, the numbers must be the same as those that were reported in part (c).
 - A response that only compares the proportion of successful (mild and severe) treatments at Clinic A to the proportion of successful (mild and severe) treatments at Clinic B should be scored I.
 - If the response in part (a-ii) concludes that Clinic B is more successful in treating allergies, then part (d) should be scored as follows:
 - **Essentially correct (E)** if the response includes the following four components:
 - (i) At least one of the two components listed for E above.
 - (ii) The response states that overall, Clinic B has a higher proportion of successfully treated allergy sufferers.
 - (iii) The response states that Clinic B has a higher proportion of successfully treated mild allergy sufferers.
 - (iv) The response states that Clinic B has a higher proportion of successfully treated severe allergy sufferers.
 - **Partially correct (P)** if the response satisfies component (i) AND two of the three remaining components (ii)–(iv) listed for E in this note.
 - If the response in part (a-ii) concludes that Clinic B is more successful, the response can be scored no higher than 3.
 - The strength of the response in part (d) should be considered if holistic scoring is necessary.
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Q6 Correct Relative Frequencies

(a-i)

	Clinic A	Clinic B
Unsuccessful treatment	$\frac{51}{139} = 0.367$	$\frac{33}{68} = 0.485$
Successful treatment	$\frac{88}{139} = 0.633$	$\frac{35}{68} = 0.515$

(c-i)

	Clinic A	Clinic B
Successful treatment - Mild	$\frac{78}{104} = 0.75$	$\frac{11}{12} = 0.917$
Successful treatment - Severe	$\frac{10}{35} = 0.286$	$\frac{24}{56} = 0.429$

(c-ii)

	Clinic A	Clinic B
Mild cases treated	$\frac{104}{139} = 0.748$	$\frac{12}{68} = 0.176$
Severe cases treated	$\frac{35}{139} = 0.252$	$\frac{56}{68} = 0.824$

NOTE: To satisfy the components in part (c), the comparison needs to be made by severity WITHIN each clinic (a vertical comparison), not Clinic A to Clinic B.

Scoring for Question 6 Each essentially correct (E) part counts as 1 point, and each partially correct (P) part counts as $\frac{1}{2}$ point.	
	Score
Complete Response	4
Substantial Response	3
Developing Response	2
Minimal Response	1
If a response is between two scores (for example, $2\frac{1}{2}$ points), use a holistic approach to decide whether to score up or down, depending on the strength of the response and quality of the communication.	